

DIWALI SALES ANALYSIS

1. Project background and description

This project analyzes a Diwali Sales dataset containing over 11,000 customer transactions to understand purchasing behavior during the festive season. The objective is to identify key customer segments, sales trends, and factors influencing revenue through exploratory data analysis.

2. Dataset Summary

- Rows: 11,251 - Columns: 15 - Key Features:
- Customer demographics (Age, Gender, State, Marital Status, Occupation)
- Purchase details (Product Category, Orders, Amount)
- Missing Data: 12 values in Amount column

3. Data Cleaning Using Python

We began with data preparation and cleaning in Python

- **Data Loading** : Imported the dataset using `pandas`
- **Initial Exploration** : Used `df.info()` to check structure and `.describe()` for summary statistics.

```
df.info()
✓ 0.0s

<class 'pandas.DataFrame'>
RangeIndex: 11251 entries, 0 to 11250
Data columns (total 15 columns):
#   Column                Non-Null Count  Dtype
---  ---                -
0   User_ID               11251 non-null  int64
1   Cust_name             11251 non-null  str
2   Product_ID           11251 non-null  str
3   Gender                11251 non-null  str
4   Age_Group             11251 non-null  str
5   Age                  11251 non-null  int64
6   Marital_Status        11251 non-null  int64
7   State                 11251 non-null  str
8   Zone                  11251 non-null  str
9   Occupation            11251 non-null  str
10  Product_Category      11251 non-null  str
11  Orders                11251 non-null  int64
12  Amount                11239 non-null  float64
13  Status                0 non-null      float64
14  unnamed1              0 non-null      float64
dtypes: float64(3), int64(4), str(8)
```

```
df.describe()
✓ 0.0s

# user_id  # age  # marital_status  # orders  # amount
count      11239.0      11239.0      11239.0      11239.0      11239.0
mean      1003003.5217546045      35.41035679330901      0.42005516505027135      2.4896343091022333      9453.610552540262
std      1716.0388257054726      12.753866054953013      0.4935894048750261      1.1149672729126838      5222.355168094685
min      1000001.0      12.0      0.0      0.0      188.0
25%      1001491.5      27.0      0.0      0.0      5443.0
50%      1003064.0      33.0      0.0      0.0      8109.0
75%      1004426.5      43.0      1.0      3.0      12675.0
max      1006040.0      92.0      1.0      4.0      23952.0

8 rows x 5 cols 10 per page < < Page 1 of 1 > >
```

- **Missing Data Handling:** Checked for null values and found 12 missing values in amount column, proceeded to delete the rows containing null values.

```
df.isnull().sum()
✓ 0.0s
```

	# 0
Product_Cate	0
Orders	0
Amount	12

```
#drop null values
df.dropna(inplace=True)
✓ 0.0s
```

- **Dropped Blank Columns:** Columns 'Status' and 'unnamed1' for found to be irrelevant to the dataset, proceeded with dropping these columns

```
#drop blank columns
df.drop(['Status', 'unnamed1'], axis = 1, inplace = True)
✓ 0.0s
```

- **Changed Column Type:** Changed Column type of 'Amount' column from float to int.

```
#change column type
df['Amount'] = df['Amount'].astype('int')
✓ 0.0s
```

```
df["Amount"].dtypes
✓ 0.0s
```

```
dtype('int64')
```

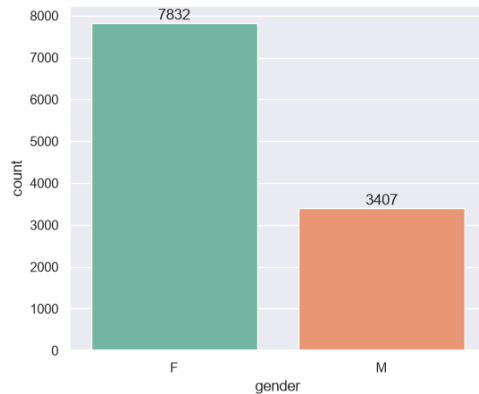
- **Column Standardization:** Renamed columns to lower case for better readability and documentation.

```
#convert column names to lowercase
df.columns = df.columns.str.lower()
✓ 0.0s
```

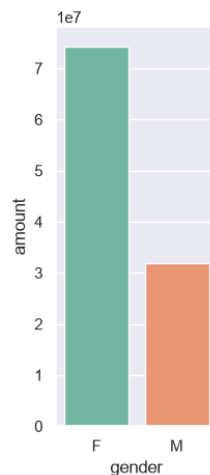
4. Data Analysis using Python

We performed structured analysis using Seaborn and Matplotlib Python libraries to answer key business questions:

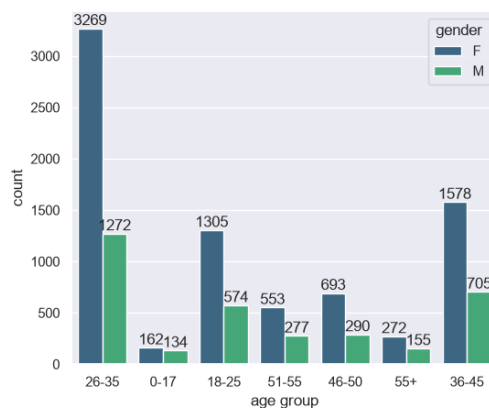
- **Gender Distribution:** Analyzed gender distribution using count plot



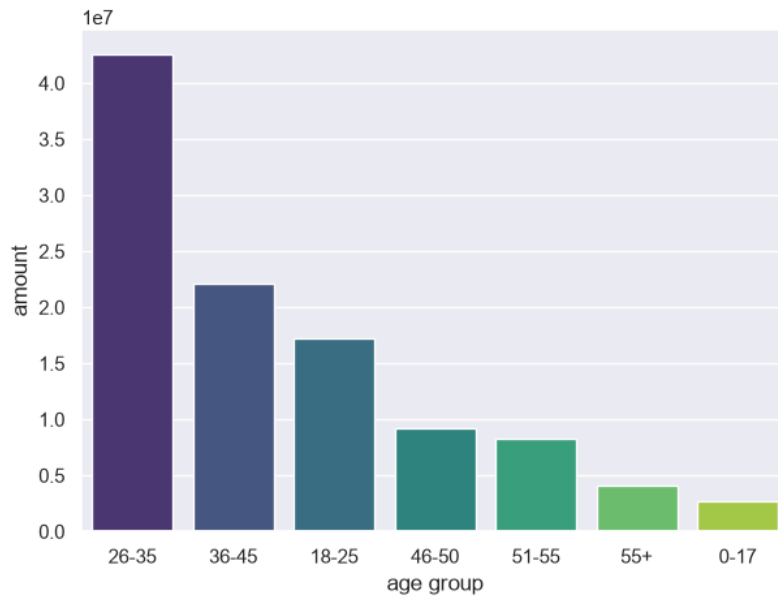
- **Total Sales Amount by Gender:** Identified which gender spent more money on buying Diwali Products



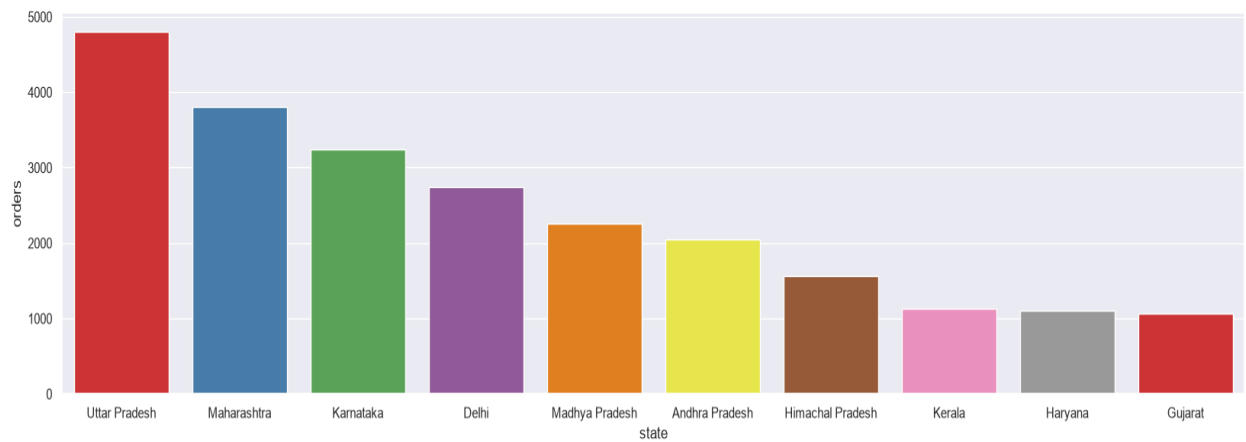
- **Customer Distribution by Age Group and Gender:** Displays the number of male and female customers across different age groups to identify demographic patterns.



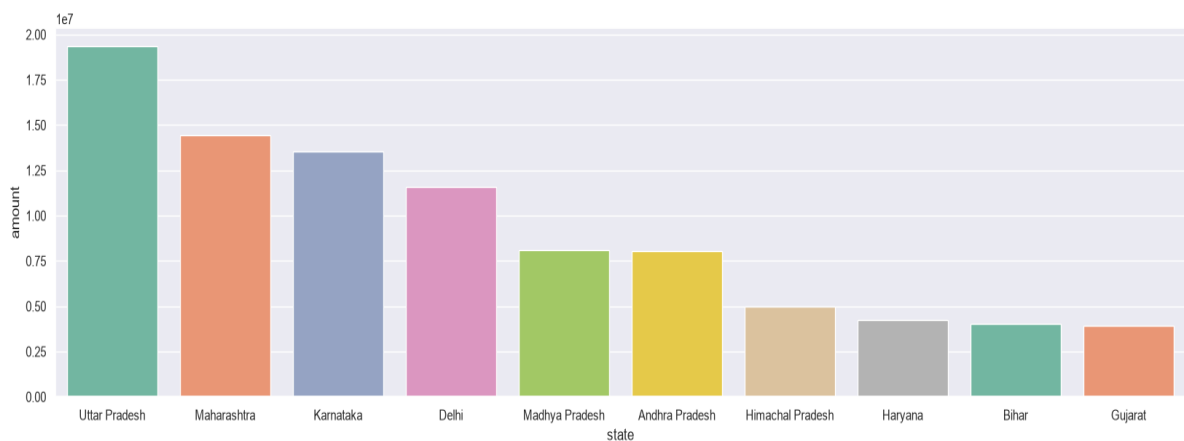
- **Total Sales Amount by Age Group:** Identified which age group spent the most money



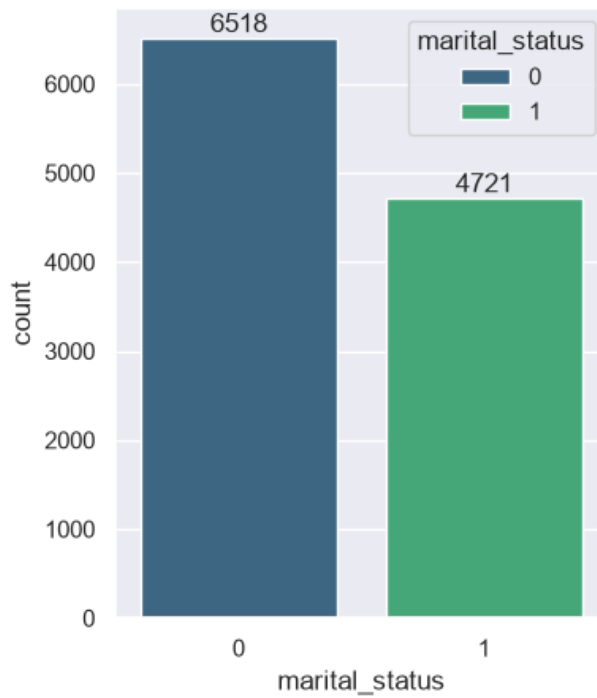
- **Total Orders by State:** Shows the distribution of customer orders across different states to identify high-demand regions.



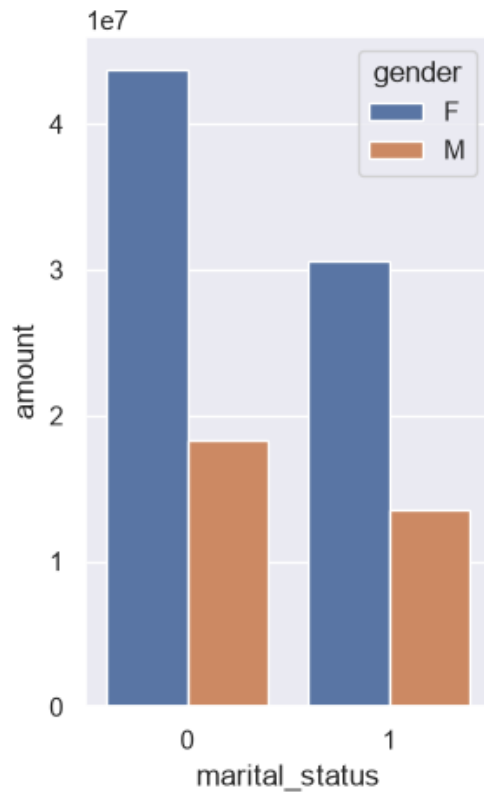
- **Total Sales Amount by State:** Displays the total sales generated across different states to identify the highest revenue-contributing regions.



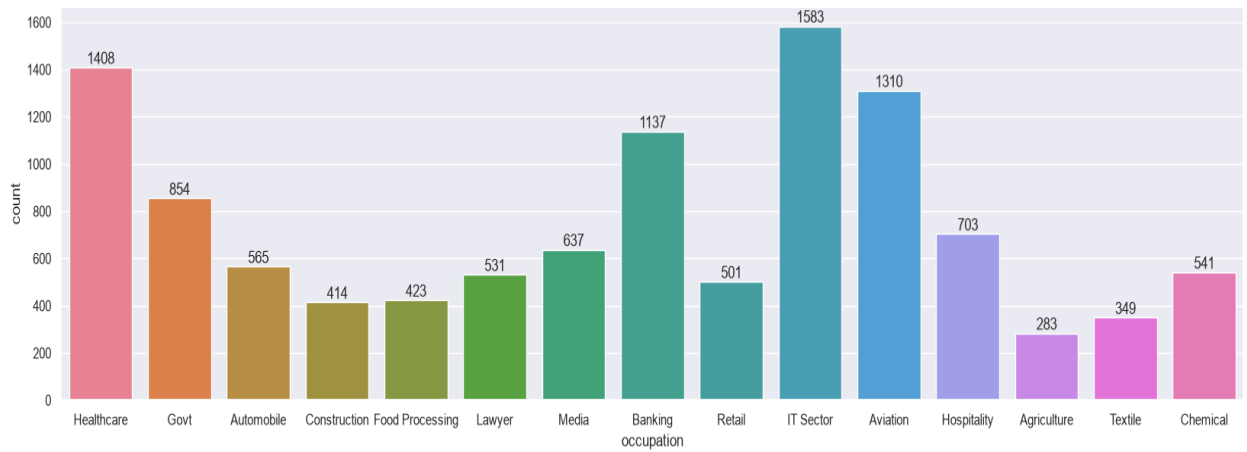
- **Customer Distribution by Marital Status:** Shows the number of customers based on their marital status to understand the demographic composition.



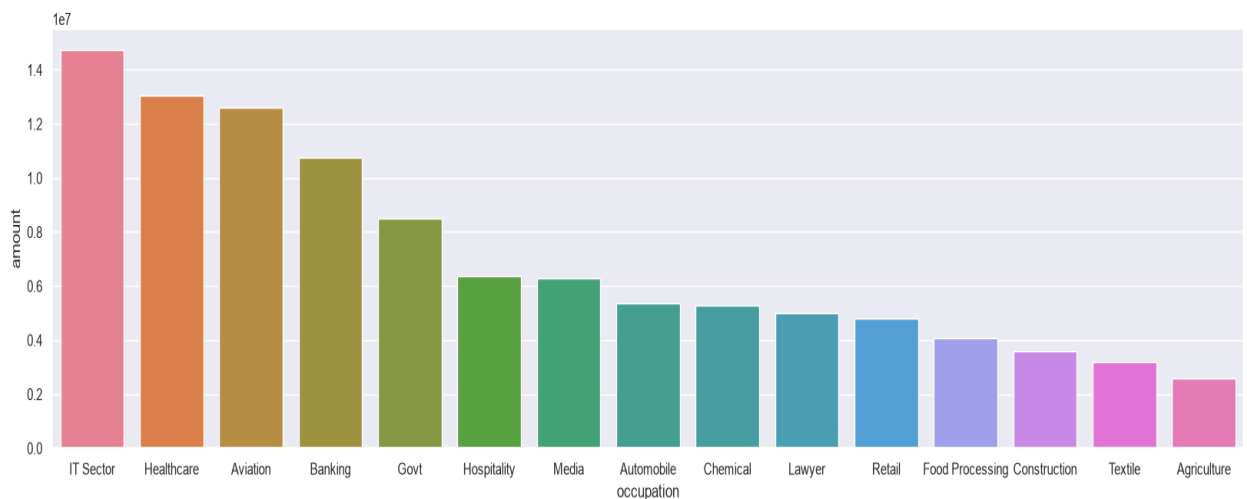
- **Total Sales Amount by Marital Status and Gender:** Compares the total sales generated by male and female customers according to marital status.



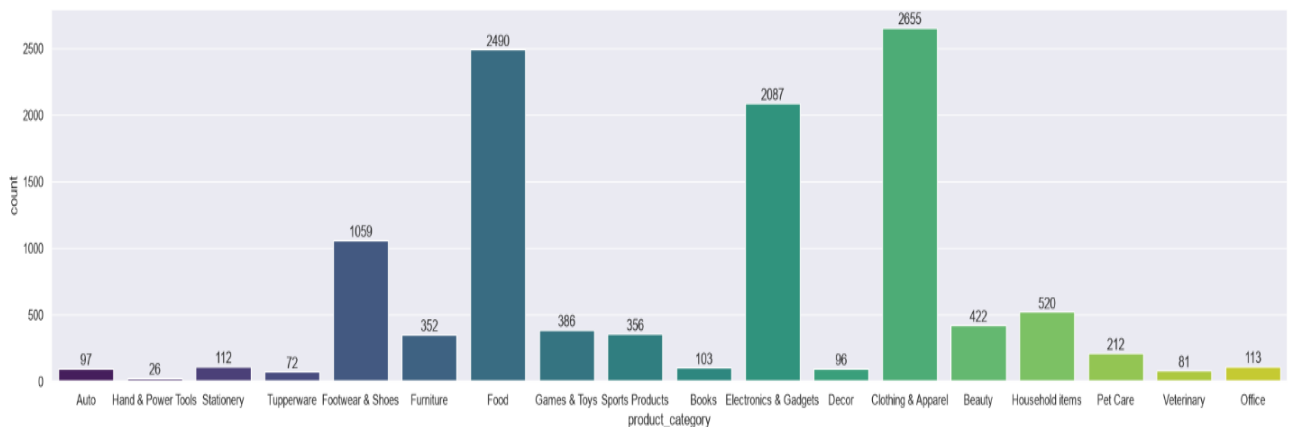
- **Customer Distribution by Occupation:** Shows the number of customers across different occupations to identify the most represented professional groups.



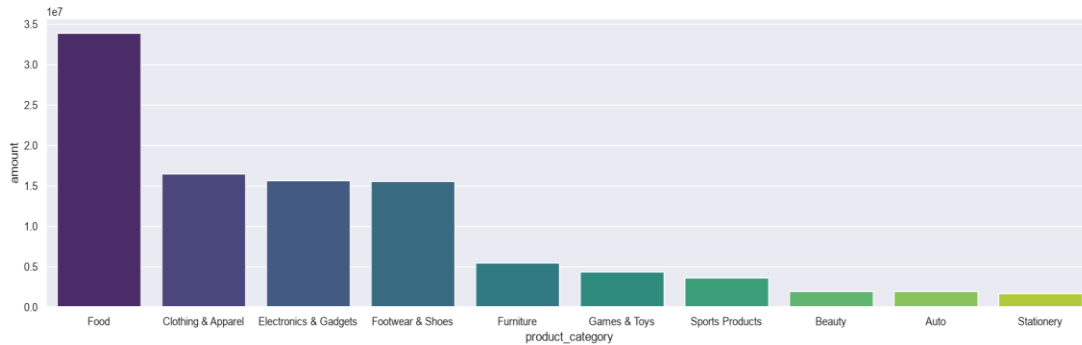
- **Total Sales Amount by Occupation:** Displays the total sales generated by customers across different occupations to identify the highest revenue-contributing professions.



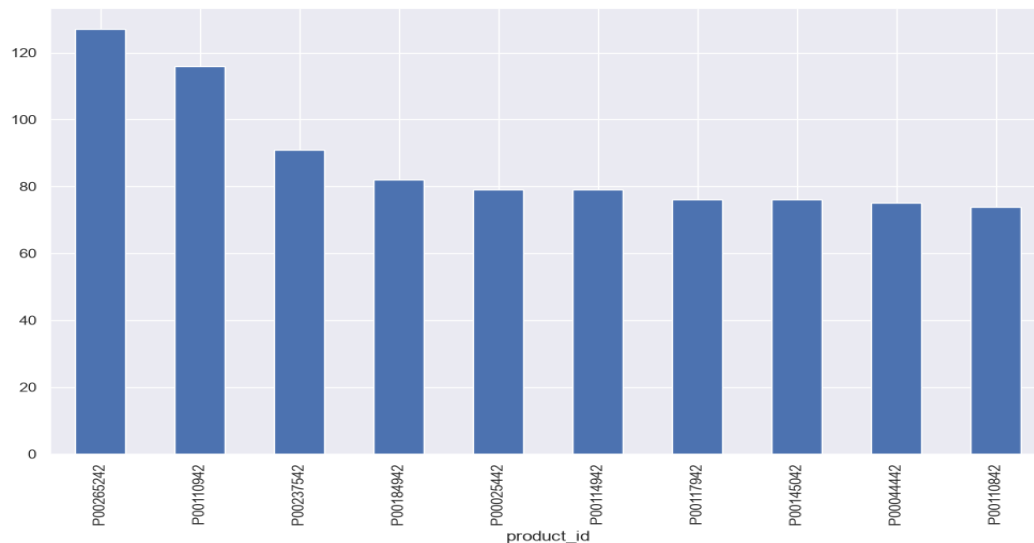
- **Customer Distribution by Product Category:** Shows the number of customer purchases across different product categories to identify the most popular products.



- **Total Sales Amount by Product Category:** Displays the total sales generated by each product category to identify the highest revenue-contributing products.



- **Top 10 Products by Order Count:** Displays the products with the highest number of orders to identify the most frequently purchased items.



5. Business Recommendations

- **Focus on female customers**, as they contribute the highest share of total sales, by designing targeted promotions and loyalty programs.
- **Prioritize customers aged 26–35**, the largest and highest-spending demographic, with personalized marketing campaigns and festive offers.
- **Increase inventory and marketing efforts in high-performing states** such as Uttar Pradesh, Maharashtra, Karnataka, and Delhi to maximize sales.
- **Target professionals in the IT, Healthcare, Aviation, and Banking sectors**, as they generate the highest revenue among occupational groups.
- **Stock and promote high-demand product categories** like Food, Clothing & Apparel, Electronics & Gadgets, and Footwear & Shoes during peak shopping seasons.
- **Introduce personalized offers based on marital status and customer demographics** to improve customer engagement and conversion rates.